

Level One Legends – Killer Robot

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Playtest Results

Design Questions

The design questions that we chose to answer with playtesting are as follows:

1. How is the game's difficulty? Are there portions that are too easy? Too difficult?

These design questions were selected to determine whether the game provides an appropriate level of difficulty for players, without being too challenging or too easy, and to determine specifically which portions of the game those are, either comprising locations in the game world, or particular aspects of the gameplay mechanics. Because our development team has spent so much time both building and testing the game, we are somewhat "blind" to how difficult the game is to a new player, so getting this feedback from playtesting is essential.

2. How is the gamefeel in terms of the two major mechanics of shooting and platforming? Is one stronger than the other?

This design question was selected to help determine the state of the gamefeel and whether either area is lagging the other and necessitating a greater degree of refinement moving forward.

3. Are there substantial glitches that impact the experience of players, and if so, to what extent do they impact the gameplay experience?

This question is included to address issues in which the game might break or cause players to become frustrated due to any technical glitches. New playtesters who are unfamiliar with the game are likely to employ different strategies or try things that we as the game's developers did not intend or try ourselves, so there is significant likelihood that playtesters may inadvertently discover glitches that are not known to us. Additionally, through playtesting we can get a better sense of the rate at which glitches occur, as well as how much frustration said glitches cause. For instance, what seems to be a minor visual glitch, while undesirable regardless, could have either a very slight or a substantial impact on the enjoyment of people playing the game, so this is valuable data to be gathered in the context of playtesting.

4. Does the game have a coherent identity with a well-defined visual aesthetic, and is that aesthetic appealing to players?

This question is important in terms of understanding both whether the game looks good and whether it has a cohesive look and the extent to which that look is consistent with what the development team intended.

5. How well does the game communicate overall goals to the player, and how well does it convey to the player the steps that need to be taken to achieve those goals?

This is a question for which playtesting is crucial; as the developers of the game, we are well aware of its goals and how the player might achieve them. Playtesting with people who have never played the game

can provide valuable insight on how well the game communicates what the player's goals should be, along with the actions that players can take to achieve those goals.

6. Are there any absent features which might be added to make the game more enjoyable?

This question aims to identify if there are any features that we might be able to add as an “easy win” to make the game more fun. While we are late in the product development timeline, if we can identify anything that might be added in this relatively short timeframe that would make the game more fun to play, then it should be considered.

7. Which aspects of the game are most in need of improvement?

This question relates to the prior gamefeel question, but with a less restricted scope. Examining this question will allow us to determine precisely where to focus our efforts over the next couple of weeks of development leading to the delivery of the final game.

8. What emergent strategies might players either discover or attempt?

This question is important to address in the context of playtesting, because we as the game developers build it with a certain implicit understanding of how the game *should* be played, and because of this, we may miss emergent strategies that could either break the game if employed, or even reveal things that could comprise interesting decisions for the players and therefore carry rewards or punishments depending on the decision that the player makes.

Approach to Testing Design Questions and Playtest Methods

The methodology used for our playtesting approach is a combination of direct observation with a spoken think-aloud, and a scripted questionnaire including short-answer questions that are focused on particular aspects of the game, Likert scale questions (disagree or agree, on a 1-5 scale), and open-ended questions that allow the playtester to provide some of their own insight or feedback outside of specifically what they are being asked about.

The actual gameplay session is to be 5-10 minutes in length, followed up by a questionnaire. This format is designed to be lengthy enough to elicit good data, but without being too long that participants become impatient with it and begin to provide less thoughtful answers or play data. Several of the design questions above are most easily addressed through directly asking the playtester the questions themselves—these include question 1 on difficulty, question 2 on gamefeel, question 6 on which features could be added to make the game more fun, and question 7 on which existing aspects of the game are most in need of improvement.

Question 4, on the game's visual aesthetic, is addressed well by not only asking the players how they liked the aesthetic, but also by asking them to qualitatively describe how they see the aesthetic, so this is asked as two separate questions within the testing apparatus.

Other questions may be addressed well through a combination of a questionnaire and observing playtester behavior and actions within the game. Design question 3, on glitches, is best answered by watching the playtester play the game, notating their spoken thoughts during the gameplay session, and then asking the playtester directly about the glitches encountered, if any, and how they affected their experience with the game.

Finally, design question 8, on emergent behavior, is best answered through observation during the think-aloud gameplay session; to the playtester it will not be clear what behavior is or is not expected to occur, but the

playtest administrator will both see emergent behavior taking place, and get insight via the think-aloud into the thought process that underlies said behavior.

PART 3: Short Answer Gamefeel Questions

SQ1: Which particular area of the game did you find most enjoyable? (Physical area, not aspect of the gameplay)

SQ2: What glitches or unintended effects, if any, stood out to you during your playtesting?

SQ3: Were there any portions of the game that you found too easy?

SQ4: Were any portions of the game too challenging?

SQ5: Which mechanics did you find most enjoyable? (Specific obstacles, enemies, platforming, etc)

SQ6: Based on the sample you have seen, describe the aesthetic of the game in a few words or a single sentence.

PART 4: Likert Scale Questions

LQ1: The game has an appealing visual style.

LQ2: The gameplay mechanics lend themselves to enjoyable play.

LQ3: The game provided me with a strong sense of what my goals were, and what steps to take to achieve those goals.

LQ4: I found myself confused while playing.

LQ5: The game had serious issues with bugs / glitches that impacted my enjoyment or ability to play.

LQ6: I found combat with enemies enjoyable in the game.

LQ7: I found platforming (jumping, navigating the course, and dodging obstacles) enjoyable in the game.

PART 5: Open Ended Questions

OQ1: What main thing(s), identifying between 1 and 3 items, specific or general, would you identify as most in need of improvement?

OQ2: Are there other particular features that you would like to see implemented? Elaborate.

OQ3: Any other thoughts on the game or project that you would like to share?

Figure 1: Excerpt from the data collection apparatus form, showing the questions asked of playtesters. Note that these also appear in the script, this excerpt is taken from the playtest data collection form, which is an Excel spreadsheet used in conjunction with the script and instruction document.

Results Summary / Analysis

Administrator	Tester Name	Age	Gender	Relationship	Death Count	Completion Time	Played Again	Visual Style	Gameplay Mechanics	Goal Communication	Clarity	Glitch Free	Combat	Platforming	Mean
Audrey Brainerd	AB1	35	Male	Brother	3	300	Yes	2	4	3	4	1	1	4	2.286
Audrey Brainerd	AB2	33	Male	Friend	1	631	Yes	2	3	1	2	4	3	4	2.714
Joshua Newsome	JN1	25	Male	Friend	0	252	No	3	2	3	2	4	1	4	2.714
Joshua Newsome	JN2	24	Female	Friend	1			4	3	2	3	4	5	3	3.429
Madeline Luna	ML1	22	Female	Friend	5	420	Yes	3	4	4	3	5	5	4	4.000
Madeline Luna	ML2	7	Female	Sister	10	1800	No	2	3	3	4	3	5	4	3.429
Mariana Zornes	MZ1	22	Male	Friend	7	612	Yes	3	4	3	3	5	3	4	3.571
Mariana Zornes	MZ2	22	Female	Friend	5	468	Yes	4	4	3	3	4	3	4	3.571
Sam Mohseni	SM1	24	Female	Girlfriend	10		No	4	3	3	4	4	1	4	3.429
Sam Mohseni	SM2	23	Male	Brother	6	660	No	3	4	5	4	4	3	4	3.857
	Mean	23.700			4.800	642.875		3.000	3.200	3.000	3.200	3.800	3.000	3.900	
	Median	23.500			5.000	540.000		3.000	3.500	3.000	3.000	4.000	3.000	4.000	
	Std Dev	7.484			3.584	491.489		0.816	1.033	1.054	0.789	1.135	1.633	0.316	
	Correlation to Mean							0.496	0.890	0.549	0.249	0.693	0.602	-0.082	

Figure 2: Image of Likert Scale question data for each playtester, showing mean, median, and std. dev as well as correlation of each feature to overall mean along with information collected on play session and playtester data (anonymized).

While the Likert scores suggested a largely glitch-free experience, two playtesters encountered glitches that caused the game to need to be reset. Glitches did not affect the gameplay experience of most participants, but where they did, they broke the game completely, meaning that these are crucially important to address. Additionally, the reporting of glitches impacting gameplay showed a strong correlation with the game's overall score performance—the player who had the lowest average assessment of the game was the one who reported the most negative experience with glitches, and within the 3 playtesters who gave the game the highest average score across categories were found both of those 5/5 scores for a glitch free experience.

6 of 10 playtesters reported that enemies were too easy to fight, owing to either the AI causing them to not fight back, or the player gun simply being too powerful in terms of its accuracy, range, and damage dealt.

2 playtesters expressed frustration with not having progress saved, either when they died and had to start the game over, or when they fell from a high area and had to climb back up from the bottom.

Features that correlated most heavily with the overall average of feature scores were as follows:

Gameplay Mechanics	0.890
Glitch Free	0.693
Combat Enjoyment	0.602
Goal Communication	0.549
Visual Style	0.496
Clarity	0.249
Platforming Enjoyment	-0.082

Among the Likert scales, six total instances of a 1 out of 5 were recorded, with 3 of those being for Combat Enjoyment, and the other three being one instance each in the categories of Gameplay Mechanics, Goal Communication, and Glitch Free Experience.

Based on the above data and the qualitative remarks provided by playtesters, key insights identified were:

1. The combat being ineffective and not having good gamefeel as a result of enemy AI being ineffectual, with playtesters remarking that enemies “didn’t fight back” and “felt like targets” and the player’s gun simply being too powerful, between having effective long range performance and very high damage output.
2. Glitches, while not highly reported, when they occurred caused major issues that impacted a player’s ability to play and enjoy the game.
3. Communication of objectives and how to achieve them is weak. Players were confused about the system by which the player character unlocks the jump ability and what the different pickup items did. While objectives were provided via text to the player, these were often missed, or they did not provide substantial meaning to the player.
4. Platforming was generally enjoyable, but players had issues with it, including being punished for trying to do something creative (2 players reported jumping on top of the piston obstacle to try to outsmart it, and instead they were punished by taking damage and not progressing, where simply going under it would at worst result in taking damage while progressing), and players frequently encountered an issue of the character getting “caught” on the edge of a platform.
5. There was a good mix in terms of which parts of the map players enjoyed. This suggests that our team has done a good job of making each area of the map interesting and varied, though there is insight into how to improve each area as well.
6. The boss fight is lacking in complexity, though this is expected with the enemies still needing substantial work to make combat effectively satisfying and challenging.
7. The understanding and description of the game’s aesthetic was generally consistent among playtesters, though the reception of the game’s visual style was still not particularly positive, with the question “the game has an appealing visual style” receiving a Likert scale score of 3, representing neither agreement nor disagreement.

Action Items / Future Work

From the data above, action items have been identified as the following, in order of most to least important:

1. Make the combat system more fun and satisfying: to make combat work better, substantial revisions to enemy AI are required. Notably, the detection range of enemies has to be larger. Playtesters reported that enemies posed a strong threat when they were near to the player, but it is trivially easy to simply shoot them from afar. To address this, we will not only increase the base detection range of the enemy AI scripts, but also add dynamic components to them that increase the size when the enemy has been shot at or when gunfire occurs in the proximity. Ideally, this might be implemented as a system in which enemy detection radii increase more based on how near the player is and how often the player fires their weapon, creating a tradeoff of shooting one enemy carrying a risk of punishment in alerting others to their location. Additionally, the player weapon must be made less effective. This will be done by causing a damage falloff based on range from the target, as well as introducing inaccuracy.
2. Improve communication to the player around objectives, what different game objects do, and what the current gamestate is. This is to be implemented by splitting the current objective system, which explains the player's in-game objectives ("Reach the next floor of the facility", "Escape the facility", etc.) with instructions around controls, such as "Press Space to Jump" into two separate systems, where players will have an objective consistently displayed in one portion of the screen, and control prompts provided in another portion of the screen as needed. Additionally, revision to the game's design language around objects such as pickup items (dropped by enemies) and exploding barrels will be made to make it clearer what each does. One player remarked that having green pickup items provide health to the player was confusing, because the health bar is pink, and not green. There is also no explanation provided to the player at this time about there being three jump parts that need to be collected to unlock the jump ability. A new interface element may be designed that shows three empty slots for the jump parts, or another system may be implemented such as the objective displaying "collect all of the jump parts" and showing a counter. There is also little indication of whether enemies are taking damage or not. To address this, we will make visual indicators of damage being taken, with enemies turning red when damaged, and sending up smoke when they are critically low on health. We will also plan to implement a health bar for the boss at the end of the level.
3. Fix all necessary glitches. This includes one glitch where the player can, in rare instances, become stuck in the ground. While this happened only once during playtesting, it resulted in the player having to restart the game, so it must be fixed for the final build. Other glitches identified included the boss having a condition by which it could become invisible, though this has been fixed already, and certain items appearing partway embedded into the ground rather than floating above it.
4. Improve platforming. Most importantly, players experienced issues with the character getting stuck on the edge of platforms when jumping onto them. In addition to fixing this, gamefeel should be refined by adjusting platform spacing to better suit the player's movement abilities, and making more challenges and puzzles around platforming.
5. Improve the game's visual appeal. While playtesters generally found that the visual language of the game was cohesive, it was also described as being very gray, empty, and bland. This will be addressed by finding more assets to include in the game as visual / decorative elements that may also be used to improve interactions, such as things to hide behind during fights, and also by improving the visuals of existing assets by providing more color and making them look more refined, in terms of animations, models, and textures.